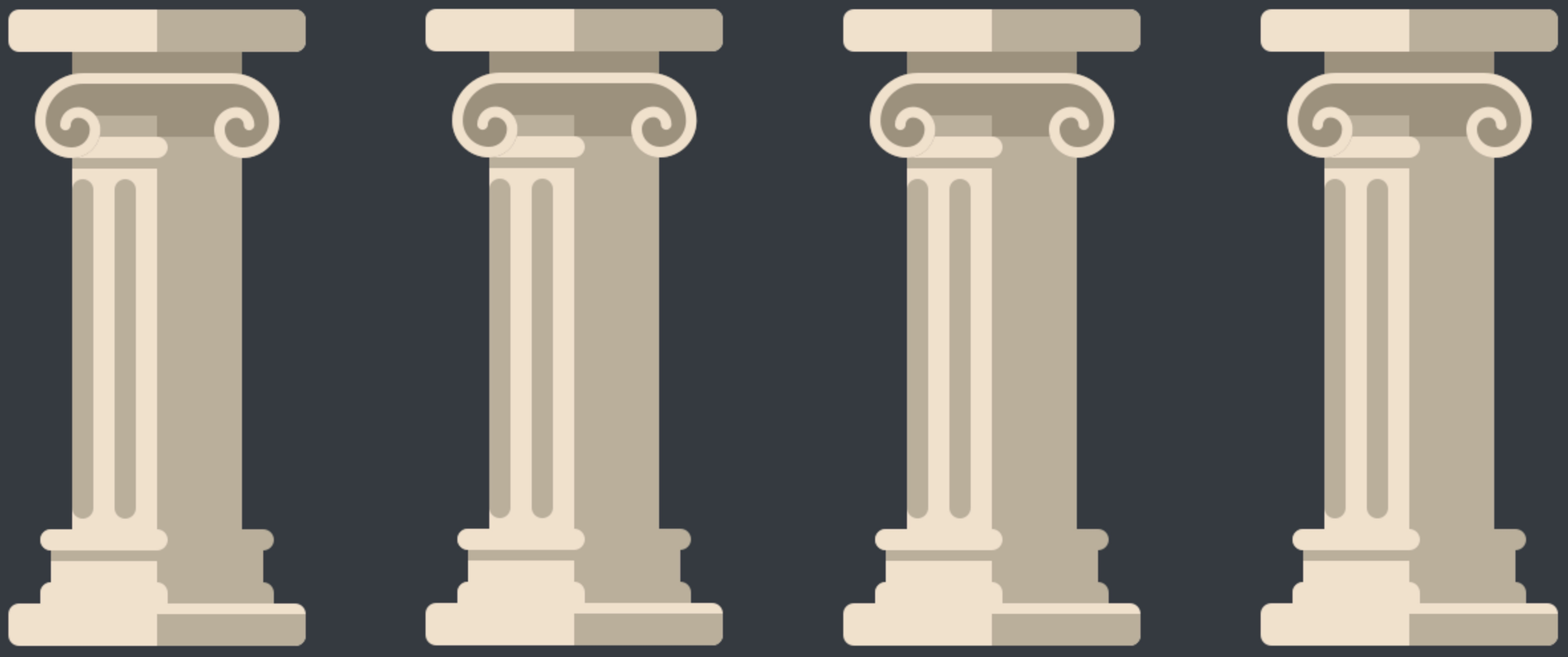
The 4 Pillars of OOP

Inheritance

Abstraction

Polymorphism

Encapsulation



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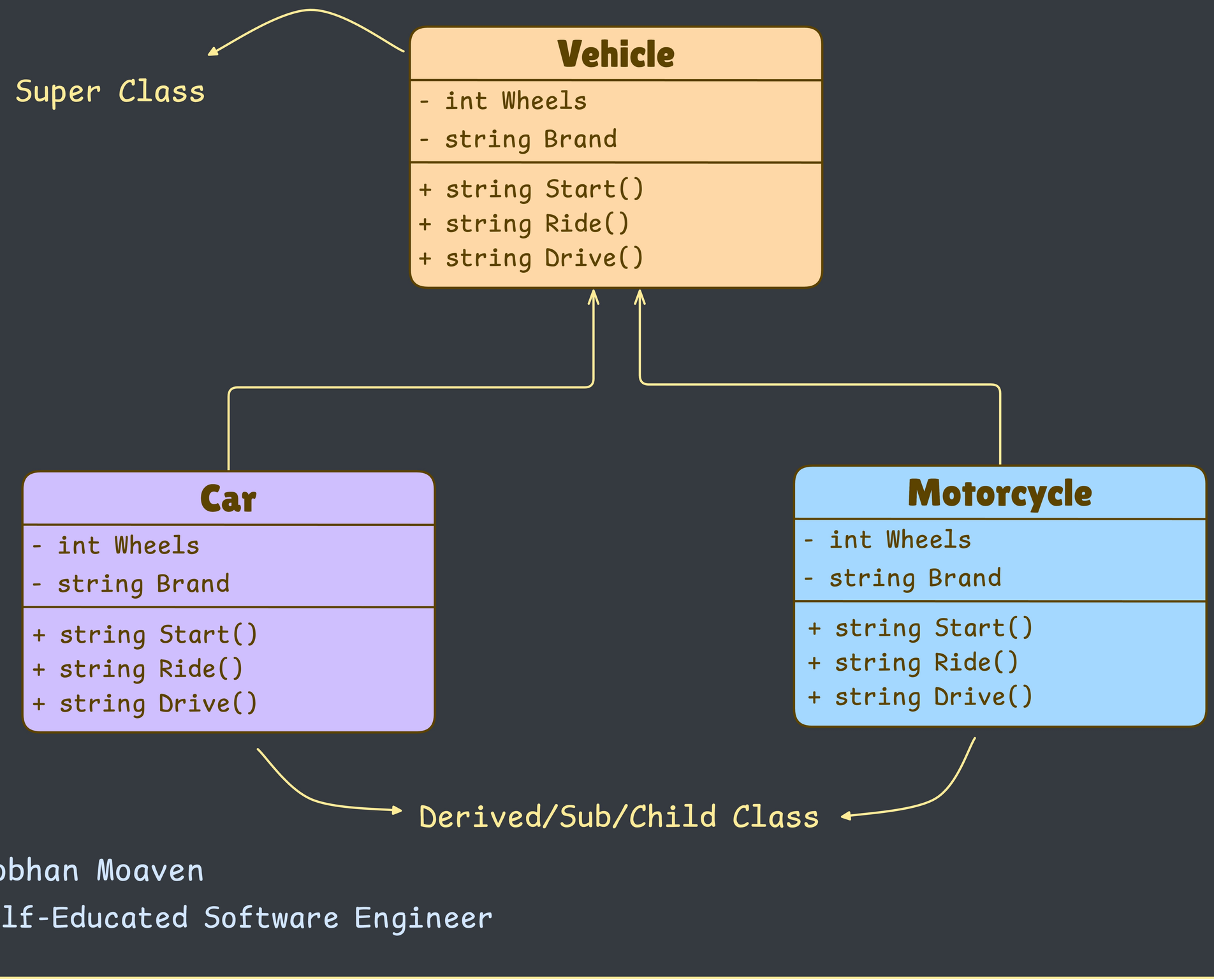
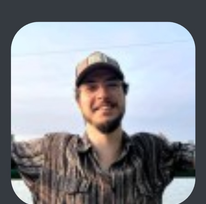
Inheritance

Inheritance refers to the ability of one class (class) to inherit properties and characteristics to another class. This increases code reusability and ensures a traditional relationship between classes.





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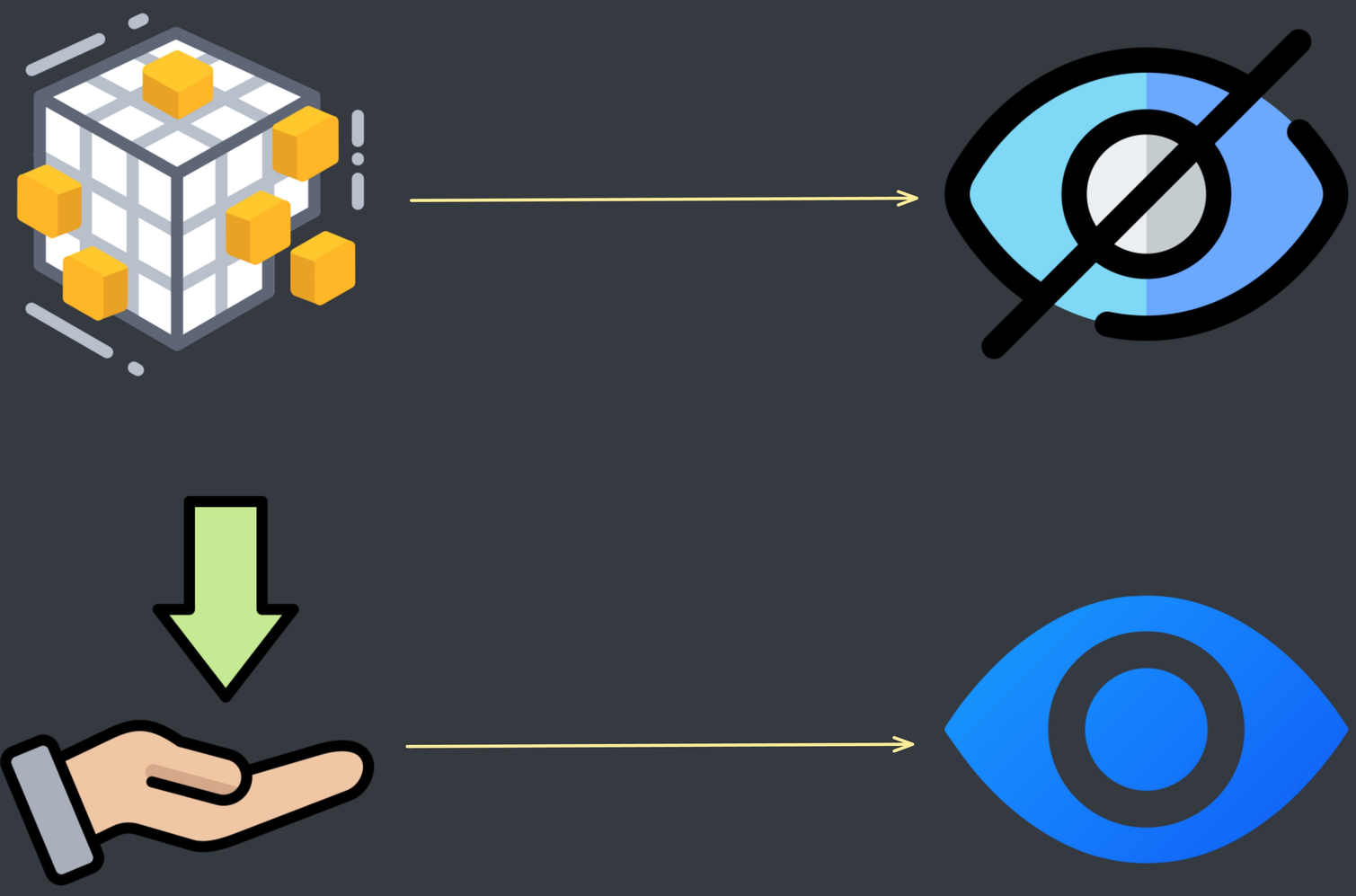
Abstraction

Abstraction aims to reduce complexity by highlighting the essential features of an object or a class, hiding unnecessary details, or neglecting irrelevant aspects.



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Encapsulation

protects the methods, data, and variables of any object from misuse by hiding them from other objects and limiting access to them.

The main advantages of Encapsulation are:

Privacy: Limits access to data and methods within a class, ensuring confidentiality by hiding internal details and protecting against unwanted external interference.

Modularity: Changes to the internal details of an object do not affect external code, promoting modularity and easier maintenance.

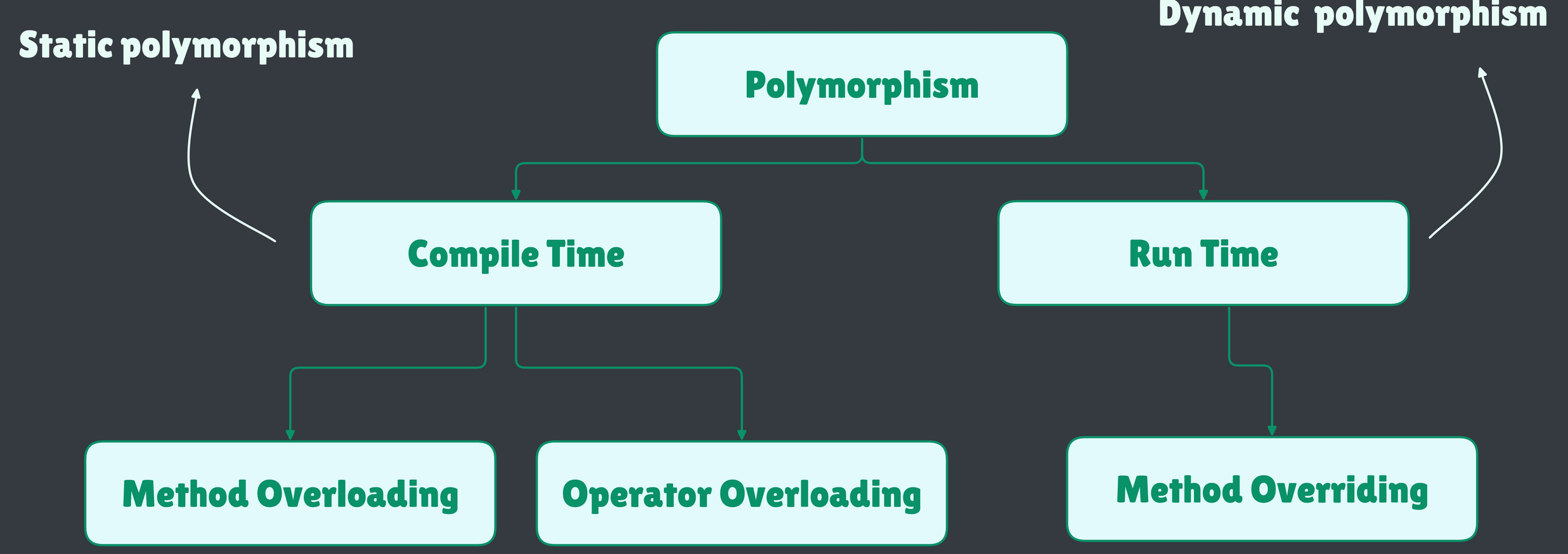
Maintainability: Encapsulation improves code maintainability by ensuring that changes to internal class details only affect methods within the class, not external code.

Encapsulation: Combines data and functionality, making the class easier to understand and encapsulated for effective use.

Sobhan Moaven
Self-Educated Software Engineer

Polymorphism

Polymorphism is the ability of objects to take on different forms, allowing the same method or operation to behave differently based on the object it is acting on.



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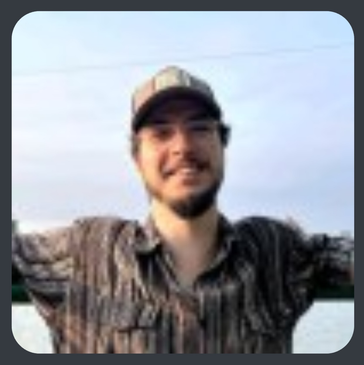

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Self-Educated Software Engineer

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Sobhan Moaven

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